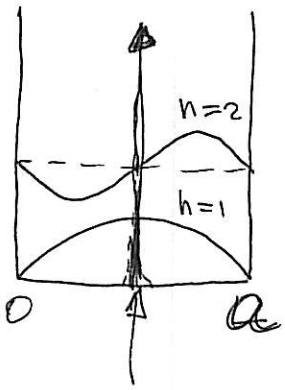
G P6.1



Infinite square well of width a

Wave functions are sine-waves

$$\psi_n^{(0)}(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x\right)$$

perturbation $H' = \alpha\, \delta(x - a/2)$ at the mid-point.

(a) $$E_n^{(1)} = \langle \psi_n^{(0)} | H' | \psi_n^{(0)} \rangle = \frac{2}{a} \int_0^a dx \sin^2\left(\frac{n\pi}{a}x\right) \alpha\, \delta\left(x - \frac{a}{2}\right) =$$

$$= \frac{2}{a}\alpha \sin^2\left(\frac{n\pi}{a}\frac{a}{2}\right) = \frac{2\alpha}{a} \sin^2\left(n\frac{\pi}{2}\right) = \begin{cases} 0, & n \in \text{even} \\ \frac{2\alpha}{a}, & n \in \text{odd} \end{cases}$$

For $n = \text{even}$ (say, $2k$) $\quad \psi^{(0)}_{n=\text{even}=2k} = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi}{a}\frac{a}{2} 2k\right) =$

$$= \sqrt{\frac{2}{a}} \sin(\pi k) = 0$$

$\Rightarrow$ even-n wave functions do not "feel" perturbation.

(b) First-order correction to the wave function

$$\psi_n^{(1)} = \sum_{m \neq n} \frac{\langle \psi_m^{(0)} | H' | \psi_n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)}$$

For the ground state $\underline{n=1}$

$\Rightarrow$ need $$\psi_1^{(1)} = \sum_{m \neq 1} \frac{\langle \psi_m^{(0)} | H' | \psi_1^{(0)} \rangle}{E_1^{(0)} - E_m^{(0)}} \psi_m^{(0)} =$$

$$= \frac{2}{a} \sum_{m \neq 1} \frac{\int_0^a dx\, \alpha\, \delta\left(x - \frac{a}{2}\right) \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{\pi x}{a}\right)}{E_1^{(0)} - E_m^{(0)}} \psi_m^{(0)} =$$

$$= \frac{2\alpha}{a} \sum_{m\neq 1} \frac{\sin\left(\frac{m\pi}{2}\right)\sin\left(\frac{\pi}{2}\right)}{\frac{\pi^2\hbar^2}{2ma^2}(1^2-m^2)} \psi_m^{(0)} = \left(\sin\left(\frac{\pi}{2}m\right) \neq 0 \text{ for } m = 3,5,7,9\ldots \text{ NOTE: } \underline{m=1} \text{ is excluded!}\right)$$

$$= \frac{4ma\alpha}{\pi^2\hbar^2} \sum_{m=3,5,7\ldots} \left\{ \frac{\sin\left(\frac{3\pi}{2}\right)\psi_3^{(0)}}{1-3^2} + \frac{\sin\left(\frac{5\pi}{2}\right)\psi_5^{(0)}}{1-5^2} + \right.$$

$$\left. + \frac{\sin\left(\frac{7\pi}{2}\right)\psi_7^{(0)}}{1-7^2} + \ldots \right\} = \frac{4ma\alpha}{\pi^2\hbar^2}\sqrt{\frac{2}{a}} \left\{ \frac{1}{8}\sin\left(\frac{3\pi}{a}x\right) + \right.$$

$$\left. - \frac{1}{24}\sin\left(\frac{5\pi}{a}x\right) + \frac{1}{48}\sin\left(\frac{7\pi}{a}x\right) + \ldots \right\} =$$

$$= \frac{m\alpha}{\pi^2\hbar^2}\sqrt{\frac{a}{2}} \left\{ \sin\left(\frac{3\pi}{a}x\right) - \frac{1}{3}\sin\left(\frac{5\pi}{a}x\right) + \frac{1}{6}\sin\left(\frac{7\pi}{a}x\right) + \ldots \right\}$$

Townsend $\boxed{P11.1}$

$H_1 = b\hat{x}^4$, $\hat{H}_0 = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$ [clearly for $x > \sqrt{\frac{m\omega^2}{2b}}$ perturbation theory should break down, see (b).]

Nonlinear (anharmonic) oscillator $\left(|\varphi_n^{(0)}\rangle = |n\rangle\right)$

(a) $E_n^{(1)} = \langle \varphi_n^{(0)} | \hat{H}_1 | \varphi_n^{(0)} \rangle =$

$$= b \langle n | \left(\sqrt{\frac{\hbar}{2m\omega}}(\hat{a}+\hat{a}^+)\right)^4 | n \rangle = \frac{b\hbar^2}{(2m\omega)^2} \langle n | (\hat{a}+\hat{a}^+)^4 | n \rangle$$

remember: $\hat{a}, \hat{a}^+$ are operators!

$$= \frac{b\hbar^2}{(2m\omega)^2} \langle n | \left(\hat{a}^2 + \hat{a}\hat{a}^+ + \hat{a}^+\hat{a} + \hat{a}^+\hat{a}^+\right)^2 | n \rangle$$

Commutator $[\hat{a}, \hat{a}^+] = 1 \Rightarrow \hat{a}\hat{a}^+ + \hat{a}^+\hat{a} = 1 + 2\hat{a}^+\hat{a}$

In $\langle n | \left(\hat{a}^2 + (\hat{a}^+)^2 + 1 + 2\hat{a}^+\hat{a}\right)^2 | n \rangle$ only

terms that "return" to $|n\rangle$ will contribute,

because $\langle n | k \rangle = 0$ for $k \neq n$. (For example $\langle n | \hat{a}^+ | n \rangle = 0$)

These are terms that contain equal number of raising $(\hat{a}^+)$ and lowering $(\hat{a})$ operators in them. Such as: $\hat{a}^2(\hat{a}^+)^2$ and its adjoint.

$$(2\hat{a}^+\hat{a}+1)^2$$

$$\Rightarrow \quad E_n^{(1)} = b\left(\frac{\hbar}{2m\omega}\right)^2 \langle n| \left[\hat{a}^2(\hat{a}^+)^2 + (\hat{a}^+)^2\hat{a}^2 + (2\hat{a}^+\hat{a}+1)^2\right]|n\rangle$$

Now use (7.32) and (7.33):

$$\hat{a}^2|n\rangle = \hat{a}\sqrt{n}\,|n-1\rangle = \sqrt{n}\sqrt{n-1}\,|n-2\rangle$$

$$(\hat{a}^+)^2|n-2\rangle = \hat{a}^+\sqrt{n-1}\,|n-1\rangle = \sqrt{n-1}\sqrt{n}\,|n\rangle$$

$$\Rightarrow \quad (\hat{a}^+)^2(\hat{a})^2|n\rangle = \sqrt{n-1}\sqrt{n}\sqrt{n}\sqrt{n-1}\,|n\rangle = n(n-1)\,|n\rangle$$

and $\hat{a}^2(\hat{a}^+)^2|n\rangle = \hat{a}^2\hat{a}^+|n+1\rangle\sqrt{n+1} = \hat{a}^2|n+2\rangle\sqrt{n+1}\sqrt{n+2} =$

$= \hat{a}|n+1\rangle\sqrt{n+2}\sqrt{n+1}\sqrt{n+2} = |n\rangle\sqrt{n+1}\sqrt{n+2}\sqrt{n+1}\sqrt{n+2} =$

$= (n+1)(n+2)\,|n\rangle$

Finally $(2\hat{a}^+\hat{a}+1)^2|n\rangle = (2\hat{n}+1)^2|n\rangle = (2n+1)^2|n\rangle$

$$\Rightarrow \quad E_n^{(1)} = b\left(\frac{\hbar}{2m\omega}\right)^2 \langle n|\, n(n-1) + (n+1)(n+2) + (2n+1)^2\,|n\rangle$$

$$= b\left(\frac{\hbar}{2m\omega}\right)^2 \underbrace{\left(n^2 - n + n + 2 + n^2 + 2n + 4n^2 + 4n + 1\right)}_{6n^2+6n+3} =$$

$$= \frac{3\hbar^2 b}{4m^2\omega^2}\left(1 + 2n + 2n^2\right).$$

(b) Compare $E_n^{(0)} = \hbar\omega\left(n+\frac{1}{2}\right)$ with $E_n^{(1)}$ for large n:

$$\frac{E_n^{(1)}}{E_n^{(0)}} = \frac{\frac{3\hbar^2 b}{4m^2\omega^2}(1+2n+2n^2)}{\hbar\omega\left(n+\frac{1}{2}\right)} \xrightarrow[n\gg 1]{} \frac{3\hbar b}{4m^2\omega^3}\,\frac{2n^2}{n} > 1 \text{ for } \boxed{n > \frac{2m^2\omega^3}{3\hbar b}}$$

P 11.2 From (11.41)

$$|\psi_n\rangle = e^{-i\frac{q|E|}{m\omega^2}\frac{\hat{p}_x}{\hbar}}|n\rangle = \left(1 - i\frac{q|E|}{m\omega^2}\frac{\hat{p}_x}{\hbar} + \dots\right)|n\rangle,$$

But (7.12): $\hat{p}_x = -i\sqrt{\frac{m\omega\hbar}{2}}\left(\hat{a} - \hat{a}^+\right)$

$$\Rightarrow |\psi_n\rangle = |n\rangle - i\frac{q|E|}{m\omega^2}\frac{(-i)}{\hbar}\sqrt{\frac{m\omega\hbar}{2}}\left(\hat{a} - \hat{a}^+\right)|n\rangle =$$

$$= |n\rangle - \frac{q|E|}{\omega}\frac{1}{\sqrt{2m\omega\hbar}}\underbrace{\left(\hat{a}|n\rangle - \hat{a}^+|n\rangle\right)}_{\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle} =$$

$$= |n\rangle - \frac{q|E|}{\omega}\frac{1}{\sqrt{2m\omega\hbar}}\left(\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle\right). \quad (1)$$

On the other hand, eq. (11.23) gives in this case

$$|\psi_n\rangle = |n\rangle + \sum_{k\neq n}\frac{\langle k|\hat{H}_1|n\rangle}{E_n^{(0)} - E_k^{(0)}}|k\rangle =$$

$$= |n\rangle + (-q|E|)\sum_{k\neq n}\sqrt{\frac{\hbar}{2m\omega}}\frac{\langle k|\hat{a} + \hat{a}^+|n\rangle}{\hbar\omega\left(n+\frac{1}{2}\right) - \hbar\omega\left(k+\frac{1}{2}\right)}|k\rangle =$$

$$= |n\rangle - \frac{q|E|}{\omega}\frac{1}{\sqrt{2m\omega\hbar}}\sum_{k\neq n}\frac{\langle k|\hat{a}|n\rangle + \langle k|\hat{a}^+|n\rangle}{n-k}|k\rangle$$

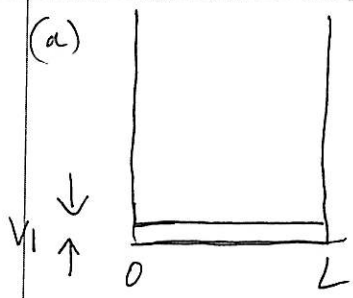
$$\left.\begin{aligned}\langle k|\hat{a}|n\rangle &= \sqrt{n}\,\langle k|n-1\rangle \Rightarrow k = n-1\\ \langle k|\hat{a}^+|n\rangle &= \sqrt{n+1}\,\langle k|n+1\rangle \Rightarrow k = n+1\end{aligned}\right\}\text{ two terms in the sum,}$$

Thus $$|\psi_n\rangle = |n\rangle - \frac{q|E|}{\omega}\frac{1}{\sqrt{2m\omega\hbar}}\left(\frac{\sqrt{n}|n-1\rangle}{n-(n-1)} + \frac{\sqrt{n+1}|n+1\rangle}{n-(n+1)}\right) =$$

$$= |n\rangle - \frac{q|E|}{\omega}\frac{1}{\sqrt{2m\omega\hbar}}\left(\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle\right) \quad (2)$$

In complete agreement with (1) that obtains by expanding the exact result eq. 11.41 to first order in perturbing el. field E.

P11.4 (a)



square well; $\hat{H}_1 = V_1$ constant change of potential energy

$$\psi_n^{(0)} = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

$$E_n^{(1)} = \langle \psi_n^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle = \underbrace{\frac{2}{L}\int_0^L dx \sin^2\left(\frac{n\pi x}{L}\right)}_{1 \text{ by normalization.}} V_1 = V_1$$

This is obviously exact result as the constant perturbation can always be absorbed into the change of the energy origin in the problem.

Thus $E_n = E_n^{(0)} + E_n^{(1)} = \frac{\hbar^2\pi^2 n^2}{2mL^2} + V_1$ (for any n).

unperturbed energy

(b) $\hat{H}_1 = \varepsilon E_1^{(0)} \frac{\hat{x}}{L}$ where $E_1^{(0)} = \frac{\hbar^2\pi^2}{2mL^2}$ (the ground state energy)

and $\varepsilon \ll 1$

Thus $E_n^{(1)} = \frac{\varepsilon E_1^{(0)}}{L} \langle \psi_n^{(0)} | \hat{x} | \psi_n^{(0)} \rangle =$

$$= \frac{\varepsilon E_1^{(0)}}{L} \frac{2}{L} \int_0^L dx\, x \sin^2\left(\frac{\pi n x}{L}\right)$$

Evaluate $I = \frac{2}{L}\int_0^L dx\, x \sin^2\left(\frac{n\pi x}{L}\right) = \left(y = \frac{\pi x}{L},\ x = \frac{L}{\pi} y\right) = \frac{2}{L}\left(\frac{L}{\pi}\right)^2 \int_0^\pi dy\, y \sin^2(ny)$

$$= \frac{2L}{\pi^2}\int_0^\pi dy\, y \sin^2(ny)$$

Do it by parts:

$$\int_0^\pi dy\, y \sin^2(ny) = -\int_0^\pi \frac{d(\cos ny)}{n}\, y \sin(ny) = \left(-\frac{1}{n}\right)\left[y \sin(ny)\cos(ny)\Big|_0^\pi - \int_0^\pi \cos(ny)\, d\big(y \sin(ny)\big)\right]$$

$= 0$ for any integer $\underline{n}$

Thus $\int_0^\pi dy\, y\, \sin^2(ny) = -\frac{1}{n}\, y \sin(n\pi)\cos(n\pi) + \frac{1}{n}\int_0^\pi \cos(ny)\cdot$

$$\cdot\left[\sin(ny)\,dy + y\cdot n\cdot\cos(ny)\,dy\right] =$$

$$= \int_0^\pi dy\, y\, \cos^2(ny) + \frac{1}{n}\underbrace{\int_0^\pi dy\, \cos(ny)\,\sin(ny)}_{-\frac{1}{n^2}\int_0^\pi d(\cos ny)\,\cos(ny) =}$$

$$= -\frac{1}{2n^2}\cos^2(ny)\Big|_0^\pi = 0 \quad \text{always.}$$

Hence we find

$$\int_0^\pi dy\, y\, \sin^2(ny) = \int_0^\pi dy\, y\, \cos^2(ny) = \int_0^\pi dy\, y\left(1-\sin^2(ny)\right)$$

$$\Rightarrow 2\int_0^\pi dy\, y\, \sin^2(ny) = \int_0^\pi dy\cdot y = \frac{\pi^2}{2}$$

$$\Rightarrow \int_0^\pi dy\, y\, \sin^2(ny) = \frac{\pi^2}{4} \quad \text{for any } \underline{n}.$$

$$\Rightarrow I = \frac{2L}{\pi^2}\cdot\frac{\pi^2}{4} = \frac{L}{2}$$

$$\Rightarrow E_n^{(1)} = \frac{\varepsilon E_1^{(0)}}{L}\cdot\frac{L}{2} = \frac{\varepsilon E_1^{(0)}}{2} = \frac{\varepsilon}{2}\,\frac{\hbar^2\pi^2}{2mL^2}$$

Thus, the energy correction is n-independent!

Note $E_n = E_n^{(0)} + E_n^{(1)} = \frac{\hbar^2\pi^2}{2mL^2}\left(n^2 + \frac{\varepsilon}{2}\right)$

so that the relative correction $\delta_n = \frac{E_n^{(0)} + E_n^{(1)}}{E_n^{(0)}} = \frac{n^2 + \frac{\varepsilon}{2}}{n^2} = 1 + \frac{\varepsilon}{2n^2}$

decreases as $\underline{n}$ increases: perturbation affects the low-energy states of the well most.

P11.5 (a) $\hat{H} \to \begin{pmatrix} E_0 + \mu|\mathcal{E}| & -A \\ -A & E_0 - \mu|\mathcal{E}| \end{pmatrix}$ in the basis of $|1\rangle - |2\rangle$ states (re-visit Ch. 4.5 for this)

$\Rightarrow$ Eigenvalue problem:

$$\begin{pmatrix} E_0 + \mu|\mathcal{E}| & -A \\ -A & E_0 - \mu|\mathcal{E}| \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = E\begin{pmatrix} a \\ b \end{pmatrix}$$

$\Rightarrow$ eigenvalues are: $\det\begin{pmatrix} E_0 + \mu|\mathcal{E}| - E & -A \\ -A & E_0 - \mu|\mathcal{E}| - E \end{pmatrix} = 0$

$$\Rightarrow (E_0 - E)^2 - \mu^2|\mathcal{E}|^2 - A^2 = 0$$

$$\Rightarrow E - E_0 = \pm\sqrt{\mu^2\mathcal{E}^2 + A^2}$$

$$\Rightarrow E = E_0 \pm \sqrt{\mu^2\mathcal{E}^2 + A^2}$$

<u>Eigenstates:</u> USE $E = E_0 \pm \sqrt{\mu^2\mathcal{E}^2 + A^2}$

$$\Rightarrow \begin{pmatrix} \mu|\mathcal{E}| \mp \sqrt{\mu^2\mathcal{E}^2 + A^2} & -A \\ -A & -\mu|\mathcal{E}| \mp \sqrt{\mu^2\mathcal{E}^2 + A^2} \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$\Rightarrow \begin{cases} \left(\mu|\mathcal{E}| \mp \sqrt{\mu^2\mathcal{E}^2 + A^2}\right) a = Ab \\ -Aa = \left(\mu|\mathcal{E}| \pm \sqrt{\mu^2\mathcal{E}^2 + A^2}\right) b \end{cases}$$

$$\Rightarrow b = \frac{-A}{\mu|\mathcal{E}| \pm \sqrt{\mu^2\mathcal{E}^2 + A^2}}\, a$$

let us simplify notations by choosing the following parametrization: $\left.\begin{aligned} \mu|\mathcal{E}| &= \rho \sin\varphi \\ A &= \rho\cos\varphi \end{aligned}\right\} \Rightarrow \sqrt{\mu^2\mathcal{E}^2 + A^2} = \rho$

$$\Rightarrow b = \frac{-\rho\cos\varphi}{\rho\sin\varphi \pm \rho}\, a = \frac{\mp\cos\varphi}{1 \pm \sin\varphi}\, a$$

Normalization: $a^2+b^2=1$

$$\Rightarrow a^2\left[1+\left(\frac{\mp\cos\varphi}{1\pm\sin\varphi}\right)^2\right]=1 \Rightarrow a^2\,\frac{1\pm 2\sin\varphi+\sin^2\varphi+\cos^2\varphi}{(1\pm\sin\varphi)^2}=1$$

$$\Rightarrow a=\frac{1\pm\sin\varphi}{\sqrt{2}\,(1\pm\sin\varphi)^{1/2}}=\sqrt{\frac{1\pm\sin\varphi}{2}}$$

$$\Rightarrow b=\frac{\mp\cos\varphi}{1\pm\sin\varphi}\sqrt{\frac{1\pm\sin\varphi}{2}}=\frac{\mp\cos\varphi}{\sqrt{2(1\pm\sin\varphi)}}$$

$\oplus$ sign for $E_+=E_0+\sqrt{\mu^2\varepsilon^2+A^2}$

$\ominus$ sign for $E_-=E_0-\sqrt{\mu^2\varepsilon^2+A^2}$

(b) First let us get $\frac{\mu|\varepsilon|}{A}\ll 1$ ~~results~~ limit from exact eigenstates above:

$$\frac{\mu|\varepsilon|}{A}=\frac{\rho\sin\varphi}{\rho\cos\varphi}=\tan\varphi\ll 1 \Rightarrow \varphi\ll 1 \Rightarrow \boxed{\frac{\mu|\varepsilon|}{A}=\varphi}$$

$\Rightarrow \cos\varphi\simeq 1$ and $\sin\varphi\simeq\varphi$

Then $$a=\frac{1}{\sqrt{2}}\left(1\pm\frac{\sin\varphi}{2}\right)=\frac{1}{\sqrt{2}}\left(1\pm\frac{\mu|\varepsilon|}{2A}\right)=\begin{cases}a_+ & +\text{ sign}\\ a_- & -\text{ sign}\end{cases}$$

$$b=\frac{1}{\sqrt{2}}\left(1\mp\frac{\sin\varphi}{2}\right)=\frac{1}{\sqrt{2}}\left(\mp 1+\frac{\mu|\varepsilon|}{2A}\right)=\begin{cases}b_+ & -\text{sign}\\ b_- & +\text{sign}\end{cases}$$

Thus $$|\psi_+\rangle=a_+|1\rangle+b_+|2\rangle=$$

$$=\frac{1}{\sqrt{2}}\left(1+\frac{\mu|\varepsilon|}{2A}\right)|1\rangle+\frac{1}{\sqrt{2}}\left(-1+\frac{\mu|\varepsilon|}{2A}\right)|2\rangle=$$

$$=\frac{1}{\sqrt{2}}\left(|1\rangle-|2\rangle\right)+\frac{\mu|\varepsilon|}{2A}\,\frac{1}{\sqrt{2}}\left(|1\rangle+|2\rangle\right)=\left(\text{USE (4.52)(4.53)}\right)$$

$$=|\mathrm{II}\rangle+\frac{\mu|\varepsilon|}{2A}|\mathrm{I}\rangle$$

Similarly $$|\psi_-\rangle=a_-|1\rangle+b_-|2\rangle=$$

$$=\frac{1}{\sqrt{2}}\left(1-\frac{\mu|\varepsilon|}{2A}\right)|1\rangle+\frac{1}{\sqrt{2}}\left(1+\frac{\mu|\varepsilon|}{2A}\right)|2\rangle=$$

$$=\frac{1}{\sqrt{2}}\left(|1\rangle+|2\rangle\right)-\frac{\mu|\varepsilon|}{2A}\,\frac{1}{\sqrt{2}}\left(|1\rangle-|2\rangle\right)=|\mathrm{I}\rangle-\frac{\mu|\varepsilon|}{2A}|\mathrm{II}\rangle$$

Now, the limit $\mu|\varepsilon| \ll A$ corresponds to eq. 4.61 (with $\omega=0$)
(that is, the basis now is $|I\rangle - |II\rangle$ one!)

$$\hat{H} \to \begin{pmatrix} E_0 - A & \mu|\varepsilon| \\ \mu|\varepsilon| & E_0 + A \end{pmatrix} = \underbrace{\begin{pmatrix} E_0 - A & 0 \\ 0 & E_0 + A \end{pmatrix}}_{\hat{H}_0 \text{ (diagonal in } |I\rangle - |II\rangle)} + \underbrace{\begin{pmatrix} 0 & \mu|\varepsilon| \\ \mu|\varepsilon| & 0 \end{pmatrix}}_{\hat{H}_1}$$

(off-diagonal: connects $|I\rangle$ to $|II\rangle$ and vice versa)

The ground state $|\varphi_-\rangle$ corresponds to $E = E_0 - A$

(11.23) with $k=II \neq n=I$

$$|\varphi_-\rangle = |I\rangle + |II\rangle \frac{\langle II|\hat{H}_1|I\rangle}{E_I^{(0)} - E_{II}^{(0)}} = |I\rangle + |II\rangle \frac{\mu|\varepsilon|}{E_0 - A - (E_0 + A)} =$$

$$= |I\rangle - \frac{\mu|\varepsilon|}{2A}|II\rangle$$

The excited state $E = E_0 + A$

(11.23) with $k=I \neq n=II$

$$|\varphi_+\rangle = |II\rangle + |I\rangle \frac{\langle I|\hat{H}_1|II\rangle}{E_{II}^{(0)} - E_I^{(0)}} =$$

$$= |II\rangle + |I\rangle \frac{\mu|\varepsilon|}{E_0 + A - (E_0 - A)} = |II\rangle + \frac{\mu|\varepsilon|}{2A}|I\rangle$$

And we finally observe that exact $|\psi_\pm\rangle$ correspond to perturbative $|\varphi_\pm\rangle$ completely!

P11.6 $s = 1/2$ particle $\hat{H} = -\hat{\mu}\cdot\vec{B} = -\frac{gq}{2mc}\hat{S}\cdot\vec{B}$

But $\vec{B} = B_0\vec{k} + B_2\vec{j}$, $B_2 \ll B_0$.

$$\Rightarrow \hat{H} = -\frac{gqB_0}{2mc}\hat{S}_z - \frac{gqB_2}{2mc}\hat{S}_y \equiv \omega_0\hat{S}_z + \omega_2\hat{S}_y$$

In the $\hat{S}_z$ basis ($|\pm z\rangle$ states)

$$\hat{H} \to \omega_0 \underbrace{\frac{\hbar}{2}\begin{pmatrix}1&0\\0&-1\end{pmatrix}}_{\text{Eq.(3.77)}} + \omega_2 \underbrace{\frac{\hbar}{2}\begin{pmatrix}0&-i\\i&0\end{pmatrix}}_{\text{Eq.(3.89)}} = \frac{\hbar}{2}\begin{pmatrix}\omega_0 & -i\omega_2\\ i\omega_2 & -\omega_0\end{pmatrix}$$

$\Rightarrow$ the eigenvalue problem $\frac{\hbar}{2}\begin{pmatrix}\omega_0 & -i\omega_2\\ i\omega_2 & -\omega_0\end{pmatrix}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = \frac{\hbar}{2}\lambda\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix}$

$$\Rightarrow \det\begin{pmatrix}\omega_0-\lambda & -i\omega_2\\ i\omega_2 & -\omega_0-\lambda\end{pmatrix} = 0 \Rightarrow \lambda^2 - \omega_0^2 - \omega_2^2 = 0$$

$$\Rightarrow \lambda = \pm\sqrt{\omega_0^2+\omega_2^2} = \pm\left|\frac{gq}{2mc}\right|\sqrt{B_0^2+B_2^2}$$

For $B_2 \ll B_0$, we have $\omega_2 \ll \omega_0 \Rightarrow$ expand in $\frac{\omega_2}{\omega_0} \ll 1$:

$$\lambda = \pm\omega_0\sqrt{1+\frac{\omega_2^2}{\omega_0^2}} = \pm\omega_0\left(1+\frac{\omega_2^2}{2\omega_0^2}\right)$$

$\Rightarrow$ the energies are $E_\pm = \frac{\hbar}{2}\lambda = \pm\frac{\hbar\omega_0}{2}\left(1+\frac{\omega_2^2}{2\omega_0^2}\right)$ (1)

<u>In perturbative calculation</u>

$$\hat{H}_0 = \frac{\hbar}{2}\begin{pmatrix}\omega_0 & 0\\ 0 & -\omega_0\end{pmatrix}, \quad \hat{H}_1 = \frac{\hbar}{2}\begin{pmatrix}0 & -i\omega_2\\ i\omega_2 & 0\end{pmatrix}$$

The eigenstates of $\hat{H}_0$ are usual $|\pm z\rangle$ states with eigenvalues $E_\pm^{(0)} = \pm\frac{\hbar}{2}\omega_0$

Corrections to these are given by (11.25) [because (11.15) give 0 as $\langle\pm|\hat{H}_1|\pm\rangle = 0$]

$$\Rightarrow E_-^{(2)} = \frac{|\langle +z|\hat{H}_1|-z\rangle|^2}{E_-^{(0)}-E_+^{(0)}} = \frac{\left(\frac{\hbar}{2}\omega_2\right)^2}{-\frac{\hbar}{2}\omega_0-\frac{\hbar}{2}\omega_0} = -\frac{\hbar}{4}\frac{\omega_2^2}{\omega_0}$$

$$E_+^{(2)} = \frac{|\langle -z|\hat{H}_1|+z\rangle|^2}{E_+^{(0)}-E_-^{(0)}} = \frac{\left(\frac{\hbar}{2}\omega_2\right)^2}{\frac{\hbar}{2}\omega_0-\left(-\frac{\hbar}{2}\omega_0\right)} = \frac{\hbar}{4}\frac{\omega_2^2}{\omega_0}$$

$$\Rightarrow E_\mp = E_\mp^{(0)} + E_\mp^{(2)} = \mp\frac{\hbar\omega_0}{2} \mp \frac{\hbar}{4}\frac{\omega_2^2}{\omega_0} = \mp\frac{\hbar\omega_0}{2}\left(1+\frac{\omega_2^2}{2\omega_0^2}\right)$$ just as in (1).